

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the `README.md` for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

1.a

Link to my NumPy tutoring session: <https://claude.ai/share/9365fd54-617b-4519-875c-ed96bf982080>

1.b **Answer:** $O(mnp)$.

Justification. The product $C = AB$ has $m \cdot p$ entries, and each operation is an inner product of length n : so each operation costs n multiply-adds. The total is therefore $mp \cdot n = O(mnp)$.

1.c Link to my einops tutoring session: <https://claude.ai/share/f6b32ebe-894a-4ff8-b7a5-2ff4666d93cf>

1.d (i) Xw , **shapes** $(n, d), (d,) \rightarrow (n,)$

```
einsum(X, w, 'n d, d -> n')
```

The single axis of w is labelled d so that it aligns with the column axis of X . Since d is absent from the output it is summed over, giving $(Xw)_i = \sum_d X_{id}w_d$. The axis n is carried through, so the result has one entry per row of X .

(ii) $XX^\top$, **shapes** $(n, d), (n, d) \rightarrow (n, n)$

```
einsum(X, X, 'i d, j d -> i j')
```

Entry $[i, j]$ is the dot product of row i with row j , that is $(XX^\top)_{ij} = \sum_d X_{id}X_{jd}$. The two row axes index *different* rows, so they were assigned different letters i and j , both of which are kept in the output. The feature axis d is shared by both operands and dropped from the output, so it is summed over.

(iii) $\text{diag}(X^\top X)$, **shapes** $(n, d), (n, d) \rightarrow (d,)$

```
einsum(X, X, 'n d, n d -> d')
```

Using the same index letter for the column axis of both inputs forces the row and column index of $X^\top X$ to be equal, which selects the diagonal entries: $y[d] += X[n, d] \cdot X[n, d]$, each entry squared and added into its own column's position on the output vector. The n axis is contracted away because it is not named in the output.

2.a

Gradient. $\nabla f(\mathbf{w}) = 2(\mathbf{w} - \mathbf{c})$.**Derivation.** Differentiation is linear, so it moves inside the sum:

$$\frac{\partial f}{\partial w_j} = \frac{\partial}{\partial w_j} \sum_{i=1}^d (w_i - c_i)^2 = \sum_{i=1}^d \frac{\partial}{\partial w_j} (w_i - c_i)^2.$$

The function is *separable*: the term indexed by i involves only w_i . Every term with $i \neq j$ is therefore a constant with respect to w_j and differentiates to zero, so only the $i = j$ term survives. Applying the chain rule to that one term, and using $\frac{d}{dw_j}(w_j - c_j) = 1$ since c_j is constant,

$$\frac{\partial f}{\partial w_j} = 2(w_j - c_j).$$

Every component has the same form, so stacking them gives

$$\nabla f(\mathbf{w}) = 2(\mathbf{w} - \mathbf{c}).$$

Evaluated at $\mathbf{w} = \mathbf{0}$.

$$\nabla f(\mathbf{0}) = 2(\mathbf{0} - \mathbf{c}) = -2\mathbf{c}.$$

2.c

Product. Each entry of C is one row of A against one column of B :

$$\begin{aligned} C_{11} &= 2 \cdot 7 + 1 \cdot 9 + 3 \cdot 1 = 26, & C_{12} &= 2 \cdot 8 + 1 \cdot 0 + 3 \cdot 2 = 22, \\ C_{21} &= 4 \cdot 7 + 5 \cdot 9 + 6 \cdot 1 = 79, & C_{22} &= 4 \cdot 8 + 5 \cdot 0 + 6 \cdot 2 = 44, \end{aligned}$$

so

$$C = AB = \begin{pmatrix} 26 & 22 \\ 79 & 44 \end{pmatrix}, \quad s = 26 + 22 + 79 + 44 = 171.$$

Gradient with respect to A . Since s sums every entry of C exactly once, $\frac{\partial s}{\partial C_{i,j}} = 1$ for all i, j , and $C_{i,j} = \sum_k A_{i,k} B_{k,j}$, so

$$\frac{\partial s}{\partial A_{i,k}} = \sum_j \frac{\partial s}{\partial C_{i,j}} \cdot B_{k,j} = \sum_j B_{k,j},$$

the row sums of B : $7 + 8 = 15$, $9 + 0 = 9$, $1 + 2 = 3$. Every row of A meets the same rows of B , so the result repeats down both rows:

$$\frac{\partial s}{\partial A} = \begin{pmatrix} 15 & 9 & 3 \\ 15 & 9 & 3 \end{pmatrix}.$$

Gradient with respect to B . By the mirrored argument,

$$\frac{\partial s}{\partial B_{k,j}} = \sum_i \frac{\partial s}{\partial C_{i,j}} \cdot A_{i,k} = \sum_i A_{i,k},$$

the column sums of A : $2 + 4 = 6$, $1 + 5 = 6$, $3 + 6 = 9$, giving

$$\frac{\partial s}{\partial B} = \begin{pmatrix} 6 & 6 \\ 6 & 6 \\ 9 & 9 \end{pmatrix}.$$

Each gradient has the same shape as the matrix it is taken with respect to, 2×3 and 3×2 respectively.

3.a

Minimizer.

$$\theta^* = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i},$$

the weighted average of the x_i .

Derivation. Differentiation is linear, so it moves inside the sum, and the chain rule applies to each term with $\frac{d}{d\theta}(\theta - x_i) = 1$ since x_i is constant:

$$f'(\theta) = \sum_{i=1}^n 2w_i(\theta - x_i).$$

Setting $f'(\theta) = 0$, cancelling the factor of 2 and splitting the sum,

$$\sum_{i=1}^n w_i \theta = \sum_{i=1}^n w_i x_i.$$

Since θ carries no index i , it factors out of the left-hand sum, giving $\theta \sum_i w_i = \sum_i w_i x_i$, and dividing by $\sum_i w_i$ yields θ^* above.

The optimum is a minimum. Expanding the first derivative gives $f'(\theta) = a\theta - c$ with $a = 2 \sum_i w_i$ and $c = 2 \sum_i w_i x_i$, so

$$f''(\theta) = 2 \sum_{i=1}^n w_i,$$

the second derivative gives a constant, and strictly positive because every $w_i > 0$. The function is therefore an upward parabola with a single stationary point, so θ^* is its global minimum.

Negative weights. Only $\sum_i w_i$ appears in f'' , so the sign of that *sum* decides the behaviour, not the sign of any individual weight. Three cases:

- $\sum_i w_i > 0$: still a genuine minimum, but θ^* may fall outside the range of the x_i . For $x = [1, 3]$ with $w = [2, -1]$, $\theta^* = -1$.
- $\sum_i w_i = 0$: the quadratic terms cancel, f reduces to a straight line and is unbounded below; the formula for θ^* divides by zero.
- $\sum_i w_i < 0$: the parabola opens downward, so θ^* is a *maximum*, $f \rightarrow -\infty$ in both directions and no minimum exists.

3.b

Transcript of my gradient descent tutoring session:

<https://claude.ai/share/ddd9eae2-8131-4c41-b06a-9924d9fd2fdc>

4.a

This model has likely encoded, contains, or potentially exacerbates bias against people of a certain gender, race, sexuality, or who have other protected characteristics. Location likely acted as substitute for the underlying income inequalities: because of residential segregation and unequal rural and urban access to credit, a postcode carries information about the applicant's race and income. The disparity in outcomes is visible in the results, namely loans are accepted from urban centres and denied to rural applicants, and the false positive rate is significantly higher for Black and male applicants. So the model reproduces existing inequalities without considering the characteristics such as race and gender directly.

4.b

Stylometry models can contain information that could be deduced about individuals that they have not consented to share. The output could likely identify authors who do not want to be identified, and texts could even be attributed to people who did not author them.

4.c

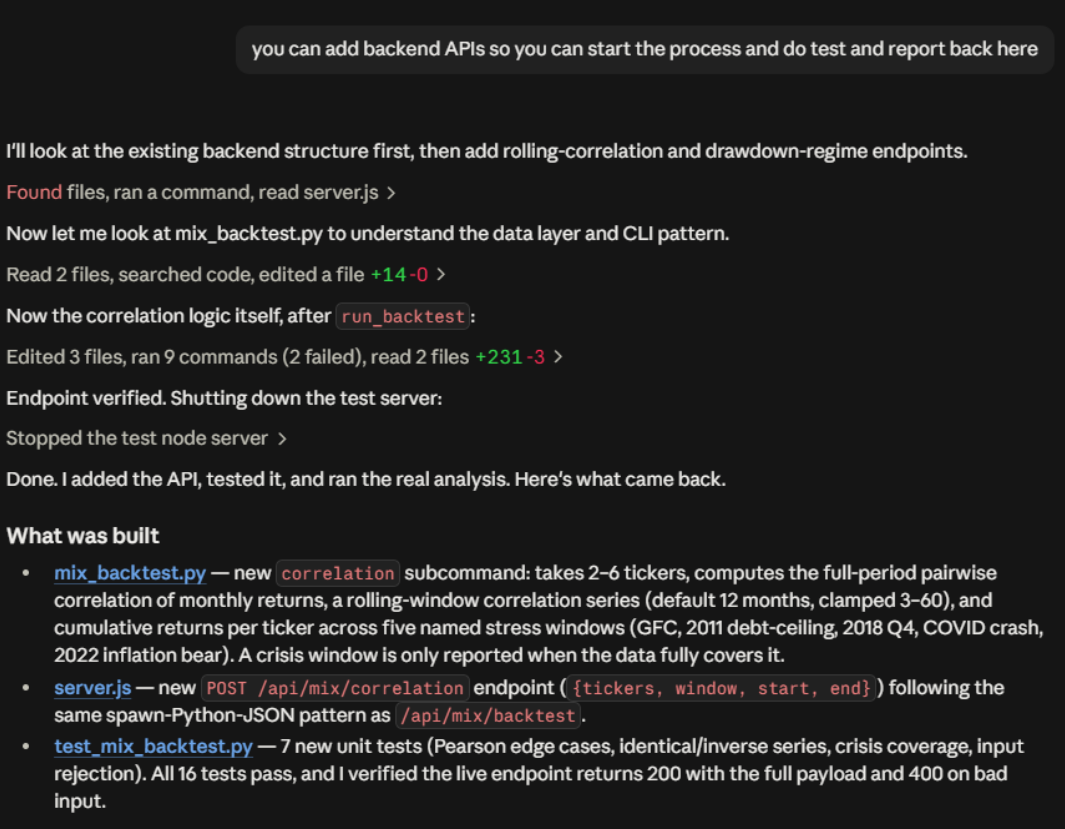
This dataset appears to contain any personally identifiable information or sensitive personally identifiable information. The celebrities depicted never gave permission for their images to be collected or used as training data, and being a public figure is not consent to become a training example. The dataset's handling of the copyrighted photos avoids redistributing protected works, but it does nothing about the underlying problem, since the faces are still assembled into a facial recognition dataset without the knowledge or agreement of the people in them.

4.d

I do not think this scenario raises an ethical concern from the guidelines. The photographs were uploaded by users who consented to their use for research, and the model is used only within the app and only for the purpose the data was collected for, so there is no repurposing beyond what those users agreed to. The images are of plants rather than people, so the dataset carries no personally identifiable information, and a species classifier has no demographic outputs through which it could encode bias against race, gender, or other protected characteristics.

5.a

The product I chose is Claude Code, from Anthropic. Claude Code was released as a CLI tool for code generation and for scanning a codebase through text prompts. Since then the tools available to it have expanded to running shell commands, editing files and creating commits and pull requests, which has vastly increased what a single prompt can do. With the arrival of agents it now supports semi autonomous sessions, where the agent is given a goal and uses the various tools to reach it, asking for approval according to the permission mode rather than for every action. Two uses stood out for me. The first is agentic development, shown in Figure 1: from one request it read an existing investment portfolio backtester, added a rolling correlation subcommand and a matching API endpoint, wrote seven unit tests, and checked that the running server answered correctly on both valid and invalid input. Figure 2 shows the same pattern building an application from scratch, creating four files and rewriting the game mod that feeds it. The second is tutoring, shown in Figure 3: while I was learning NumPy for this course it generated practice questions on indexing and shape, gave hints, and held the answers back until I attempted them.



```

you can add backend APIs so you can start the process and do test and report back here

I'll look at the existing backend structure first, then add rolling-correlation and drawdown-regime endpoints.
Found files, ran a command, read server.js >
Now let me look at mix_backtest.py to understand the data layer and CLI pattern.
Read 2 files, searched code, edited a file +14 -0 >
Now the correlation logic itself, after run_backtest:
Edited 3 files, ran 9 commands (2 failed), read 2 files +231 -3 >
Endpoint verified. Shutting down the test server:
Stopped the test node server >
Done. I added the API, tested it, and ran the real analysis. Here's what came back.

What was built
• mix_backtest.py — new correlation subcommand: takes 2-6 tickers, computes the full-period pairwise correlation of monthly returns, a rolling-window correlation series (default 12 months, clamped 3-60), and cumulative returns per ticker across five named stress windows (GFC, 2011 debt-ceiling, 2018 Q4, COVID crash, 2022 inflation bear). A crisis window is only reported when the data fully covers it.
• server.js — new POST /api/mix/correlation endpoint ( {tickers, window, start, end} ) following the same spawn-Python-JSON pattern as /api/mix/backtest .
• test_mix_backtest.py — 7 new unit tests (Pearson edge cases, identical/inverse series, crisis coverage, input rejection). All 16 tests pass, and I verified the live endpoint returns 200 with the full payload and 400 on bad input.

```

Figure 1: Agentic development on an existing codebase: reading the backend, adding an endpoint, writing unit tests and verifying the live responses.



Figure 2: The same agentic pattern building an application from scratch.

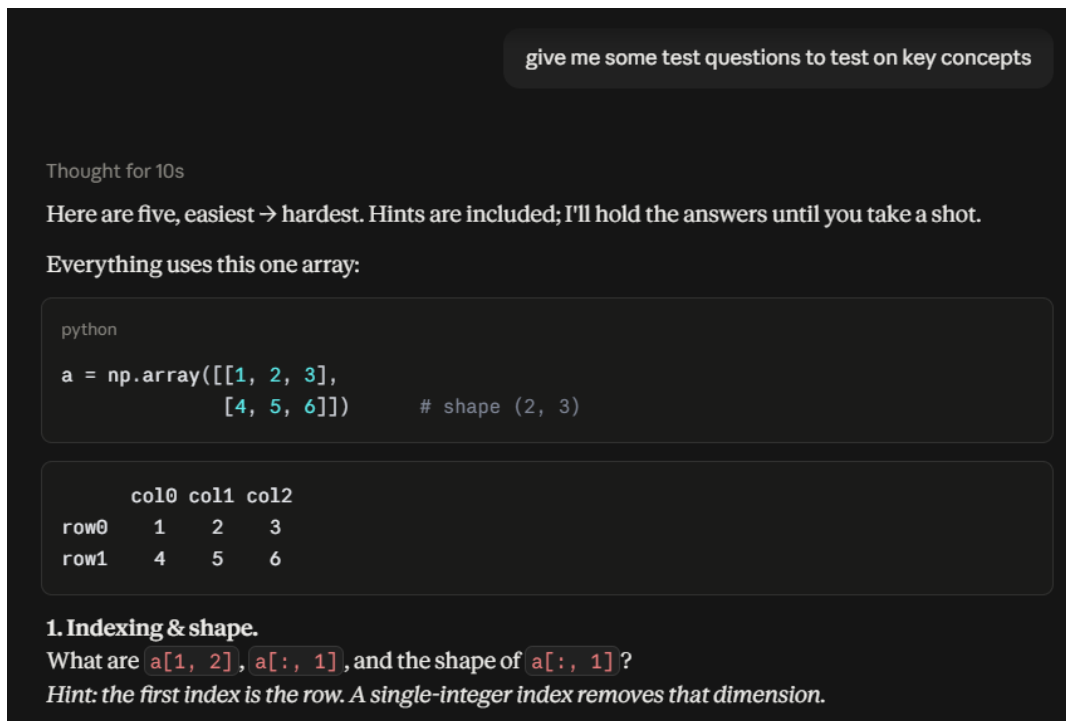


Figure 3: Tutoring use: practice questions on NumPy indexing and shape, with hints and answers withheld.

5.b

Table 5 was temporarily blocked for all access after the U.S. government issued an export control order that also covered Mythos 5 (12 June 2026). Anthropic complied but arguing the jailbreak cited revealed only minor and already known vulnerabilities, and the episode created controversy over how access to frontier models is regulated. Mythos 5 was later restored to about a hundred vetted U.S. organisations while wider and international access stayed restricted. Anthropic also reported that Claude designed protein binders against fifteen targets and succeeded on fourteen, at hit rates above the industry norm (18 August 2026), which shows the company expanding into real world impact beyond an AI chatbot. Most notably, research on a global workspace probes the inner thinking steps of the model and opens up the black box of LLM reasoning, showing that fabrication or deception can be detected in representations that never reach the output (6 July 2026). The company has since extended free and discounted access to verified scientists (27 August 2026).

5.c

Company mission. Anthropic states: “We believe AI will have a vast impact on the world. Anthropic is dedicated to building systems that people can rely on and generating research about the opportunities and risks of AI.” (anthropic.com/company)

Product goal. Claude Code is described as excelling “at both routine development tasks like bug fixes and testing, as well as transformative work like refactors and feature implementation that require deep codebase understanding.”

Relationship. I believe the product aligns with the mission, since Claude Code is a system that applies AI beyond simple chat, with impact on code generation such as tests and codebase understanding. Those are also tasks where the output has to be reliable, because the result is committed into a real codebase. The tension is that a mission built around researching the risks of AI also produces a tool that writes files and runs shell commands on its own, which carries the capability of reaching external systems.

Alignment. What I observed matched the stated goal. In Figure 1 it reported that nine commands ran and two failed instead of hiding the failures, and it tested the error response as well as the success response rather than only the case that looks good. I did see one mistake, where a debugging line was pasted into a function whose variable did not exist and immediately errored, but that gained the model nothing, so I would call it an ordinary error rather than reward hacking. Anthropic handles misalignment mainly through the permission system and environmental control, which asks for approval before the tool acts and lets the user set how much it may do unattended.

5.d

Pricing and access. Claude is sold on individual and enterprise plans (claude.com/pricing). The free tier costs nothing and gives basic chat access with no Claude Code access. Pro at \$23 per month billed annually, or \$28 billed monthly, is the entry point for Claude Code, and the Max tiers raise usage limits five and twenty times starting at \$140 per month. Access is tiered rather than equitable, since the tool is unavailable to anyone who will not pay a subscription. Claude for Education (claude.com/solutions/education) exists but publishes no pricing, and a recent programme gives verified scientists free standard seats for a year and premium seats at \$15 per month ([announcement](#)), though both require belonging to a recognised institution or profession.

Non-financial barriers. Claude Code has required terminal access, which excludes users who would benefit from it but do not work at a command line, though it has recently been added to the Claude desktop app for easier access. Capability is also gated separately from price: the top tier Mythos model is only available to selected companies and not to the public, so the most capable system cannot be reached at any consumer price. Region availability and an account in good standing are further conditions that money does not resolve.

An alternative pricing model. A per product subscription would fit better than one plan covering everything. Users who focus on coding could subscribe to Claude Code alone without paying for chat access they do not use, and chat only users could be charged less, since answering a question needs far less compute than searching and editing across a codebase. For users this means paying for the product they actually want and a lower entry price for the lighter one, which widens access. For the company it prices each product closer to what it costs to serve and makes the cheaper tier attractive to new users, but it splits the subscriber base, complicates the billing and gives up the revenue from users who currently pay one price for everything and use only part of it.

Computational and environmental cost. Anthropic has no formal statements on environmental cost, especially for data centres: no emissions figures, no per query energy and nothing about the centres themselves, so users cannot weigh

the environmental cost of a session against its benefit. Given the rising price of computing chips and memory, reflected in the quarterly revenue of \$96.2 billion reported by NVIDIA, more than double a year earlier ([Q2 FY2027 results](#)), the cost of compute will only rise and cause an eventually higher barrier to access. Newer open weight models such as Kimi K3 from Moonshot AI add competitive pressure that should benefit the overall landscape, though competition alone does nothing to make the environmental cost visible.